TOWARDS EXTENDING THE AHLSWEDE–KHACHATRIAN THEOREM TO CROSS t-INTERSECTING FAMILIES

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Abstract. Ahlswede and Khachatrian's diametric theorem is a weighted version of their complete intersection theorem, which is itself a well known extension of the t-intersecting Erdős-Ko-Rado theorem. The complete intersection theorem says that the maximum size of a family of subsets of $[n] = \{1, \ldots, n\}$, every pair of which intersects in at least t elements, is the size of certain trivially intersecting families proposed by Frankl. We address a cross intersecting version of their diametric theorem.

Two families $\mathcal{A}$ and $\mathcal{B}$ of subsets of $[n]$ are *cross t-intersecting* if for every $A \in \mathcal{A}$ and $B \in \mathcal{B}$, A and B intersect in at least t elements. The p-weight of a k element subset A of $[n]$ is $p^k(1-p)^{n-k}$, and the weight of a family $\mathcal{A}$ is the sum of the weights of its sets. The weight of a pair of families is the product of the weights of the families.

The maximum p-weight of a t-intersecting family depends on the value of p. Ahlswede and Khachatrian showed that for p in the range $[\frac{r}{t+2r-1}, \frac{r+1}{t+2r+1}]$, the maximum p-weight of a t-intersecting family is that of the family $\mathcal{F}_r^t$ consisting of all subsets of $[n]$ containing at least $t+r$ elements of the set $[t+2r]$.

In a previous paper we showed a cross t-intersecting version of this for large t in the case that $r = 0$. In this paper, we do the same in the case that $r = 1$. We show that for p in the range $[\frac{1}{t+1}, \frac{2}{t+3}]$ the maximum p-weight of a cross t-intersecting pair of families, for $t \geq 200$, is achieved when both families are $\mathcal{F}_1^t$. Further, we show that except at the endpoints of this range, this is, up to isomorphism, the only pair of t-intersecting families achieving this weight.

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1. Introduction

Let $[n] := \{1, 2, \ldots, n\}$ and let $\binom{[n]}{k}$ be the family of all k-subsets of $[n]$. For a positive integer t, the family $\mathcal{A} \subset 2^{[n]}$ is called *t-intersecting* if, for each $A, A' \in \mathcal{A}$, we have $|A \cap A'| \geq t$. Erdős, Ko, and Rado proved in [4] that, for each k and t, there exists $n_0 = n_0(k,t)$ such that if $n \geq n_0$ and a family of k-element subsets $\mathcal{A} \subset \binom{[n]}{k}$ is t-intersecting, then $|\mathcal{A}| \leq \binom{n-t}{k-t}$ with equality holding if and only if there is some $T \in \binom{[n]}{t}$ such that $\mathcal{A} = \{A \in \binom{n}{k} : T \subset A\}$. The exact bound $n_0(k,t) = (t+1)(k-t+1)$ was established by Frankl [5], where he introduced the random walk method, and independently by Wilson [8], where he used a linear programming bound due to Delsarte.

Frankl also considered the case when $n < (t+1)(k-t+1)$. He defined t-intersecting families $\mathcal{F}_i^t$ by

$$\mathcal{F}_i^t = \mathcal{F}_i^t(n) = \Big\{F \subset [n] : \big|F \cap [t+2i]\big| \geq t+i\Big\},$$

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and conjectured that if $\mathcal{A} \subset \binom{[n]}{k}$ is t-intersecting, then

$$|\mathcal{A}| \leq \max_i \left|\mathcal{F}_i^t \cap \binom{[n]}{k}\right|.$$

This conjecture was partially proved by Frankl and Füredi in [6], and was finally settled by Ahlswede and Khachatrian in the affirmative in [1] and [2]. This result, now known as the complete intersection theorem, is one of the most important results in extremal set theory.

Ahlswede and Khachatrian also obtained the p-weight version of their complete intersection theorem in [3], which they call the diametric theorem. To state their result, we let p be a real number with $0 < p < 1$, and let $q := 1 - p$. For a family $\mathcal{F} \subset 2^{[n]}$, the *$p$-weight of $\mathcal{F}$* is defined by

$$\mu_p(\mathcal{F}) := \sum_{F \in \mathcal{F}} p^{|F|} q^{n-|F|}.$$

Ahlswede and Khachatrian showed that if $\mathcal{F} \subset 2^{[n]}$ is t-intersecting, then

$$\mu_p(\mathcal{F}) \leq \max_i \mu_p(\mathcal{F}_i^t). \tag{1}$$

Comparing $\mu_p(\mathcal{F}_i^t)$ and $\mu_p(\mathcal{F}_{i+1}^t)$, it can be shown that $\max_i \mu_p(\mathcal{F}_i^t) = \mu_p(\mathcal{F}_r^t)$ if and only if

$$\frac{r}{t+2r-1} \leq p \leq \frac{r+1}{t+2r+1}. \tag{2}$$

All values of $p > 0$ fall into this range for some r, smaller p yield larger r.

For a positive integer t, the families $\mathcal{A}, \mathcal{B} \subset 2^{[n]}$ are called *cross t-intersecting* if, for each $A \in \mathcal{A}$ and $B \in \mathcal{B}$, we have $|A \cap B| \geq t$. We consider an extension of (1) to cross t-intersecting families.

Conjecture 1. *If $\mathcal{A} \subset 2^{[n]}$ and $\mathcal{B} \subset 2^{[n]}$ are cross t-intersecting, then where r is such that p satisfies* (2)*,*

$$\mu_p(\mathcal{A})\mu_p(\mathcal{B}) \leq \mu_p(\mathcal{F}_r^t)^2.$$

With Frankl, in [7], we verified the $r = 0$ case of the above conjecture for $t \geq 14$. In this paper we verify the $r = 1$ case of the conjecture for $t \geq 200$. This result is perhaps the first result concerning cross intersecting families, where optimal structures are different from the so-called trivial structure $\mathcal{F}_0^t$. To state our main result we need one more definition. Two families $\mathcal{G}_1, \mathcal{G}_2 \in 2^{[n]}$ are *isomorphic*, denoted by $\mathcal{G}_1 \cong \mathcal{G}_2$, if there is a permutation σ on $[n]$ such that $\mathcal{G}_1 = \Big\{\{\sigma(k) : k \in G\} \ : \ G \in \mathcal{G}_2\Big\}$.

Theorem 1. *Let n and t be integers with $n \geq t \geq 200$, and let p be such that $\frac{1}{t+1} \leq p \leq \frac{2}{t+3}$. If $\mathcal{A} \subset 2^{[n]}$ and $\mathcal{B} \subset 2^{[n]}$ are cross t-intersecting, then*

$$\mu_p(\mathcal{A})\mu_p(\mathcal{B}) \leq \big(\mu_p(\mathcal{F}_1^t)\big)^2 = \Big((t+2)p^{t+1}q + p^{t+2}\Big)^2. \tag{3}$$

Moreover, equality holds if and only if one of the following holds:

(i) $\mathcal{A} = \mathcal{B} \cong \mathcal{F}_0^t$ *and* $p = \dfrac{1}{t+1}$,

(ii) $\mathcal{A} = \mathcal{B} \cong \mathcal{F}_1^t$ *and* $\dfrac{1}{t+1} \leq p \leq \dfrac{2}{t+3}$,

(iii) $\mathcal{A} = \mathcal{B} \cong \mathcal{F}_2^t$ *and* $p = \dfrac{2}{t+3}$.

Remark that we do not attempt to optimize the range of t. The parts requiring t to be around 200 are (7) and (25).

Organization: In Section 2 we introduce some standard definitions and techniques, and state some useful results from [7]. In Section 3 we make some quick reductions and setup parameters for the families $\mathcal{A}$ and $\mathcal{B}$ by which we break the proof down into cases.

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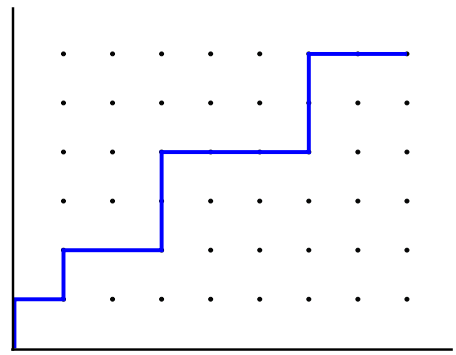

FIGURE 1. The walk $F = \{1, 3, 6, 7, 11, 12\} \subset [14]$

In particular, we introduce a pair of parameters (s, s'), with $0 \le s' \le s$, which effectively measures the difference between our families $\mathcal{A}$ and $\mathcal{B}$ and the optimal families $\mathcal{F}_0^t$, $\mathcal{F}_1^t$ or $\mathcal{F}_2^t$. When (s, s') is one of $(0,0), (1,1), (2,2)$, or $(1,0)$, then $\mathcal{A}$ and $\mathcal{B}$ will be, or will be very close to, one of these families. In this case we have to look closely at the structure of our families, and compare them with the optimal families directly. This will be done in Section 5.

The remaining cases are dealt with in Section 4. When (s, s') is not one of the above four values, then $\mathcal{A}$ and $\mathcal{B}$ are very different from the optimal families, so we can expect them to have relatively small weight. This seems as though it should make computation easier, but there is an added difficulty in that we can no longer compute their weight relative to the optimal families, rather we must compute these weights directly. That said, if s is big, a fairly crude estimation of the weight will suffice, and these cases are done in Subsection 4.1. For the intermediate values of s we define functions related to the weight of the families and show that these are monotonic in various ranges of (s, s'). Doing so we reduce the remaining cases to the final four cases of (s, s') given above. This, is done in Subsection 4.2.

This overall approach is based on the paper [7], but the monotonicity ideas used in Subsection 4.2 are new. We feel that such ideas will be necessary in proving Conjecture 1 for larger values of r.

2. Preliminaries

2.1. **Subset vs. walk on a two-dimensional grid.** It is useful to regard a set $F \subset [n]$ as a walk starting at the origin $(0,0)$ of the two-dimensional grid $\mathbb{Z}^2$ as follows. If $i \in F$, then the i-th step is *up* from (x, y) to $(x, y+1)$. Otherwise, the i-th step is *right* from (x, y) to $(x+1, y)$. For simplicity, we refer to $F \subset [n]$ as a set or a walk. See Figure 1 for an example.

Let $\mathcal{F}^\ell$ be the family of all walks that hit the line $y = x + \ell$; that is, let

$$\mathcal{F}^\ell = \Big\{ F \subset [n] : \big|F \cap [j]\big| \ge \frac{j+\ell}{2} \text{ for some } j \Big\}.$$

Partition the family $\mathcal{F}^\ell$ into the following three subfamilies:

$$\tilde{\mathcal{F}}^\ell := \big\{ F \in \mathcal{F}^\ell : F \text{ hits } y = x + \ell + 1 \big\},$$
$$\dot{\mathcal{F}}^\ell := \big\{ F \in \mathcal{F}^\ell : F \text{ hits } y = x + \ell \text{ exactly once, but does not hit } y = x + \ell + 1 \big\},$$
$$\ddot{\mathcal{F}}^\ell := \big\{ F \in \mathcal{F}^\ell : F \text{ hits } y = x + \ell \text{ at least twice, but does not hit } y = x + \ell + 1 \big\}.$$

So we can write

$$\mathcal{F}^\ell = \tilde{\mathcal{F}}^\ell \sqcup \dot{\mathcal{F}}^\ell \sqcup \ddot{\mathcal{F}}^\ell.$$

The following lemmas hold.

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Lemma 2 ([7], Lemma 2.2 (i,iii)). *For any positive integer ℓ, we have the following, where $\alpha = p/q$.*

(i) $\mu_p(\mathcal{F}^\ell) \le \alpha^\ell$ *and* $\mu_p(\tilde{\mathcal{F}}^\ell) \le \alpha^{\ell+1}$

(ii) $\mu_p(\ddot{\mathcal{F}}^\ell) \le \alpha^{\ell+1}$

Lemma 3 ([7], Lemma 2.2 (ii)). *Let $\mathcal{F} \subset 2^{[n]}$ and let ℓ be a positive integer. For every $\epsilon > 0$, there exists an n_0 such that if $n \ge n_0$ and no walk in $\mathcal{F}$ hits the line $y = x + \ell$, then*

$$\mu_p(\mathcal{F}) < 1 - \alpha^\ell + \epsilon.$$

2.2. **Inclusion maximal and shifted families.** A family $\mathcal{F} \subset 2^{[n]}$ is called *inclusion maximal* if $F \in \mathcal{F}$ and $F \subset F'$ imply $F' \in \mathcal{F}$.

Fact 4. *If $\mathcal{A}, \mathcal{B}$ are cross t-intersecting families in $2^{[n]}$, then there are inclusion maximal cross t-intersecting families $\mathcal{A}', \mathcal{B}' \in 2^{[n]}$ such that $\mathcal{A} \subset \mathcal{A}'$ and $\mathcal{B} \subset \mathcal{B}'$.*

For $F \subset [n]$ and $i, j \in [n]$, let

$$s_{ij}(F) := \begin{cases} (F \setminus \{j\}) \cup \{i\} & \text{if } F \cap \{i,j\} = \{j\} \text{ and } (F \setminus \{j\}) \cup \{i\} \notin \mathcal{F}, \\ F & \text{otherwise.} \end{cases}$$

Then, for $\mathcal{F} \subset 2^{[n]}$, let

$$s_{ij}(\mathcal{F}) := \{s_{ij}(F) : F \in \mathcal{F}\}.$$

A family $\mathcal{F}$ is called *shifted* if $s_{ij}(\mathcal{F}) = \mathcal{F}$ for all $1 \le i < j \le n$. Here we list some basic properties concerning shifting operations.

Lemma 5 ([7], Lemma 2.3). *Let $1 \le i < j \le n$ and let $\mathcal{F}, \mathcal{G} \subset 2^{[n]}$.*

(i) *Shifting operations preserve the p-weight of a family, that is, $\mu_p(s_{ij}(\mathcal{G})) = \mu_p(\mathcal{G})$.*

(ii) *If $\mathcal{F}$ and $\mathcal{G}$ in $2^{[n]}$ are cross t-intersecting families, then $s_{ij}(\mathcal{F})$ and $s_{ij}(\mathcal{G})$ are cross t-intersecting families as well.*

(iii) *For a pair of families we can always obtain a pair of shifted families by repeatedly shifting families simultaneously finitely many times.*

The following lemma, which mimics a proposition in [1], that is in turn based on an idea from [5], was stated in [7], but its proof was only sketched. We prove it now for all values of r, though we only need it for $r = 0, 1$, and 2.

Lemma 6. *Let $t \ge 2$ and let $\mathcal{A}, \mathcal{B} \subset 2^{[n]}$ be cross t-intersecting families. If $s_{ij}(\mathcal{A}) = s_{ij}(\mathcal{B}) = \mathcal{F}_r^t$, then $\mathcal{A} = \mathcal{B} \cong \mathcal{F}_r^t$.*

Proof. First we remark that if $\mathcal{A} \cong \mathcal{F}_r^t$, then $\mathcal{B} = \mathcal{A}$. Further, if $i, j \in [t+2r]$ or $i, j \notin [t+2r]$, then $\mathcal{A} = \mathcal{F}_r^t$ and we are done. So without loss of generality we may assume that $i = t + 2r$ and $j = n$. Define two subfamilies $\mathcal{A}_1$ and $\mathcal{A}_2$ of $\mathcal{A}$ by

$$\mathcal{A}_1 = \{A \in \mathcal{A} : |A| = t + r,\ i \notin A,\ j \in A,\ (A \cup \{i\}) \setminus \{j\} \notin \mathcal{A}\},$$
$$\mathcal{A}_2 = \{A \in \mathcal{A} : |A| = t + r,\ i \in A,\ j \notin A,\ (A \setminus \{i\}) \cup \{j\} \notin \mathcal{A}\}.$$

Since $s_{ij}(\mathcal{A}) = \mathcal{F}_r^t$, we have $|A \cap [t + 2r - 1]| = t + r - 1$ for all $A \in \mathcal{A}_1 \cup \mathcal{A}_2$. If $\mathcal{A}_1 = \emptyset$, then $\mathcal{A} = \mathcal{F}_r^t$. If $\mathcal{A}_2 = \emptyset$, then $\mathcal{A} = \{A \subset [n] : |A \cap ([i-1] \cup \{j\})| \ge t + r\} \cong \mathcal{F}_r^t$. So we may assume that $\mathcal{A}_1 \ne \emptyset$ and $\mathcal{A}_2 \ne \emptyset$.

Let $\mathcal{H} = \binom{[t+2r-1]}{t+r-1}$. Then for every $H \in \mathcal{H}$ we have $H \cup \{j\} \in \mathcal{A}_1$ or $H \cup \{i\} \in \mathcal{A}_2$ (but not both). Thus we can identify $\mathcal{H}$ with $\mathcal{A}_1 \sqcup \mathcal{A}_2$. We also define $\mathcal{B}_1$ and $\mathcal{B}_2$ in the same manner. Let $\mathcal{H}'$ be a copy of $\mathcal{H}$, and identify $\mathcal{H}'$ with $\mathcal{B}_1 \sqcup \mathcal{B}_2$.

Now we define a bipartite graph G on $V(G) = \mathcal{H} \sqcup \mathcal{H}'$, by letting $\{H, H'\}$ be an edge, for $H \in \mathcal{H}$ and $H' \in \mathcal{H}'$, if $|H \cap H'| = t - 1$. We claim that G is a connected graph. Indeed, the graph G_0

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defined on $\mathcal{H}$ by letting $\{H_1, H_2\}$ be an edge if $|H_1 \cap H_2| = t-1$, is Kneser's graph; and this is connected and non-bipartite for $t > 1$. If G is not connected, then each connected component is isomorphic to G_0, which contradicts the fact that G_0 is not bipartite. This shows that G is connected.

Therefore, there is a path from $A \in \mathcal{A}_1$ to $B \in \mathcal{B}_2$ in G, and on this path there is an edge $\{A_1, B_2\}$ where $A_1 \in \mathcal{A}_1$, $B_2 \in \mathcal{B}_2$, or an edge $\{A_2, B_1\}$ where $A_2 \in \mathcal{A}_2$, $B_1 \in \mathcal{B}_1$. But then $|A_1 \cap B_2| = t-1$ or $|A_2 \cap B_1| = t-1$, which contradicts the fact that $\mathcal{A}$ and $\mathcal{B}$ are cross t-intersecting. □

Fact 4 and Lemmas 5 allow us to assume that $\mathcal{A}$ and $\mathcal{B}$ are inclusion maximal and shifted in proving the inequalities in Theorem 1. Lemma 6, allows us to extend this assumption to the uniqueness results in the case of equality in the Theorem. We record this as the following assumption.

Assumption 7. *$\mathcal{A}$ and $\mathcal{B}$ are inclusion maximal and shifted.*

3. Setup for proof of Theorem 1

Recall that n and t are integers with $n \geq t \geq 200$, and p is a real number with $\frac{1}{t+1} \leq p \leq \frac{2}{t+3}$. Set $q = 1-p$ and $\alpha = p/q$. The following holds.

Lemma 8 ([7], Lemma 2.12)**.** *Let $f(n)$ be the maximum of $\mu_p(\mathcal{A})\mu_p(\mathcal{B})$ over all pairs $\mathcal{A}$ and $\mathcal{B}$ of cross t-intersecting families in $2^{[n]}$. Then, $f(n) \geq f(n+1)$.*

We may therefore assume that n is arbitrarily large.

For $\mathcal{F} \subset 2^{[n]}$, let $\lambda(\mathcal{F})$ be the maximum integer $\lambda \geq 0$ such that all walks in $\mathcal{F}$ hit the line $y = x + \lambda$. Let $u = \lambda(\mathcal{A})$ and $v = \lambda(\mathcal{B})$. The following holds.

Lemma 9 ([7], Lemma 2.11(ii))**.** *If $\mathcal{A}$ and $\mathcal{B}$ are shifted, inclusion maximal, cross t-intersecting families in $2^{[n]}$, then $\lambda(A) + \lambda(B) \geq 2t$.*

Therefore, we assume that $u + v \geq 2t$. If $u + v \geq 2t+1$, then Lemma 2 gives that

$$\mu_p(\mathcal{A})\mu_p(\mathcal{B}) \leq \mu_p(\mathcal{F}^u)\mu_p(\mathcal{F}^v) \leq \alpha^u \alpha^v \leq \alpha^{2t+1}.$$

One can check that $\alpha^{2t+1} < 0.99\big(\mu_p(\mathcal{F}_1^t)\big)^2$ for $t \geq 26$. Indeed,

$$\frac{\big(\mu_p(\mathcal{F}_1^t)\big)^2}{\alpha^{2t+1}} \geq (t+2)^2 pq^3 (q^t)^2 \geq (t+2)^2 pq^3 \frac{1}{e^4},$$

where the second inequality follows from

$$\frac{1}{e^2} < \left(1 - \frac{2}{t+3}\right)^{t+3} < \left(\frac{t+1}{t+3}\right)^t \leq q^t \leq \left(\frac{t}{t+1}\right)^t \leq 0.5. \tag{4}$$

Since pq^3 is increasing in p for $p \leq 0.25$, we have that

$$\frac{\big(\mu_p(\mathcal{F}_1^t)\big)^2}{\alpha^{2t+1}} \geq \frac{(t+2)^2 t^3}{e^4 (t+1)^4} > 1.02,$$

where the last inequality holds for $t \geq 26$.

Therefore, we assume that

$$u + v = 2t.$$

Without loss of generality, let

$$u \leq v.$$

Note that $\mathcal{A} \subset \mathcal{F}^u$. So $\mathcal{A}$ is partitioned as $\mathcal{A} = \tilde{\mathcal{A}} \sqcup \dot{\mathcal{A}} \sqcup \ddot{\mathcal{A}}$, where

$$\tilde{\mathcal{A}} := \mathcal{A} \cap \tilde{\mathcal{F}}^u,\ \dot{\mathcal{A}} := \mathcal{A} \cap \dot{\mathcal{F}}^u,\ \text{ and } \ddot{\mathcal{A}} := \mathcal{A} \cap \ddot{\mathcal{F}}^u.$$

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Similarly, we have that $\mathcal{B} = \tilde{\mathcal{B}} \sqcup \dot{\mathcal{B}} \sqcup \ddot{\mathcal{B}}$, where

$$\tilde{\mathcal{B}} := \mathcal{B} \cap \tilde{\mathcal{F}}^v,\ \dot{\mathcal{B}} := \mathcal{B} \cap \dot{\mathcal{F}}^v,\ \text{ and }\ \ddot{\mathcal{B}} := \mathcal{B} \cap \ddot{\mathcal{F}}^v.$$

If $\dot{\mathcal{A}} = \emptyset$, then $\mathcal{A} = \ddot{\mathcal{A}} \cup \tilde{\mathcal{A}}$, and hence,

$$\mu_p(\mathcal{A}) = \mu_p(\tilde{\mathcal{A}}) + \mu_p(\ddot{\mathcal{A}}) \leq \mu_p(\tilde{\mathcal{F}}^u) + \mu_p(\ddot{\mathcal{F}}^u) \leq \alpha^{u+1} + \alpha^{u+1},$$

where the last inequality follows from Lemma 2. Thus, we have that

$$\mu_p(\mathcal{A})\mu_p(\mathcal{B}) \leq (\alpha^{u+1} + \alpha^{u+1})\alpha^v \leq 2\alpha^{2t+1} < 0.99\left(\mu_p(\mathcal{F}_1^t)\right)^2,$$

where the last inequality holds for $t \geq 110$. Similarly, we have that if $\dot{\mathcal{B}} = \emptyset$, then, for $t \geq 110$,

$$\mu_p(\mathcal{A})\mu_p(\mathcal{B}) < 0.99\left(\mu_p(\mathcal{F}_1^t)\right)^2.$$

So (3) holds if $\dot{\mathcal{A}} = \emptyset$ or $\dot{\mathcal{B}} = \emptyset$.

We may therefore assume that $\dot{\mathcal{A}} \neq \emptyset$ and $\dot{\mathcal{B}} \neq \emptyset$. Recall that

$$\mathcal{F}_i^\ell = \Big\{F \subset [n] : \big|F \cap [\ell + 2i]\big| \geq \ell + i\Big\}.$$

That is, $\mathcal{F}_i^\ell$ is the family of walks hitting $(i, i+k)$ for some $k \geq \ell$. Note that as $\dot{\mathcal{A}}$ and $\dot{\mathcal{B}}$ are non-empty, there exist non-negative integers s and s' such that $\dot{\mathcal{A}} \cap \mathcal{F}_s^u \neq \emptyset$ and $\dot{\mathcal{B}} \cap \mathcal{F}_{s'}^v \neq \emptyset$. The next lemma tells us that such s and s' are unique. Its statement has been modified, but it is essentially Lemma 3.2 of [7].

Lemma 10 ([7], Lemma 3.2)**.** *Suppose that $\dot{\mathcal{A}} \neq \emptyset$ and $\dot{\mathcal{B}} \neq \emptyset$. Then, there exist unique non-negative integers s and s' such that*

$$\mathcal{A}_s := \dot{\mathcal{A}} \sqcup \ddot{\mathcal{A}} \subset \mathcal{F}_s^u \quad \textit{and} \quad \mathcal{B}_{s'} := \dot{\mathcal{B}} \sqcup \ddot{\mathcal{B}} \subset \mathcal{F}_{s'}^v.$$

Moreover, $s - s' = (v-u)/2$. In particular, $s \geq s'$.

Here, we record our **setup**.

- $\mathcal{A}$ and $\mathcal{B}$ are shifted maximal cross t-intersecting families.
- n may be assumed to be arbitrarily large.
- $q = 1 - p$ and $\alpha = p/q$.
- $t \geq 200$, and $\frac{1}{t+1} \leq p \leq \frac{2}{t+3}$, so $\frac{t+1}{t+3} \leq q \leq \frac{t}{t+1}$.
- $u + v = 2t$ and $1 \leq u \leq t \leq v \leq 2t$.
- $s \geq s' \geq 0$ and $s - s' = (v-u)/2$.
- $u = t - s + s'$ and $v = t + s - s'$.
- $\mathcal{A} = \dot{\mathcal{A}} \sqcup \ddot{\mathcal{A}} \sqcup \tilde{\mathcal{A}} \subset \mathcal{F}^u$ and $\mathcal{B} = \dot{\mathcal{B}} \sqcup \ddot{\mathcal{B}} \sqcup \tilde{\mathcal{B}} \subset \mathcal{F}^v$.
- $\dot{\mathcal{A}} \neq \emptyset$, $\dot{\mathcal{B}} \neq \emptyset$, $\dot{\mathcal{A}} \sqcup \ddot{\mathcal{A}} \subset \mathcal{F}_s^u$, and $\dot{\mathcal{B}} \sqcup \ddot{\mathcal{B}} \subset \mathcal{F}_{s'}^v$.

4. Almost all cases

The rest of the proof is broken down into cases based on the value of (s, s'). We first deal with the cases with $s \geq 10$. Then we spend the rest of the section reducing the remaining cases to the four final cases which will be proved in Section 5.

4.1. Large values of s.

Let

$$\bar{\mathcal{F}}_i^\ell := (\dot{\mathcal{F}}^\ell \cup \ddot{\mathcal{F}}^\ell) \cap \mathcal{F}_i^\ell.$$

We use the following key estimation from [7].

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Claim 11 ([7], Claim 3.3)**.** *There is an integer n_0 such that if $n > n_0$, then*

$$\mu_p(\tilde{\mathcal{F}}^\ell \cup \bar{\mathcal{F}}_i^\ell) < f(\ell, i, p) \cdot 1.001,$$

where

$$f(\ell, i, p) := \alpha^{\ell+1} + \binom{\ell+2i}{i} \frac{\ell+1}{\ell+i+1} p^{\ell+i} q^i (1-\alpha). \tag{5}$$

Now, since $\mathcal{A} \subset \tilde{\mathcal{F}}^u \cup \bar{\mathcal{F}}_s^u$ and $\mathcal{B} \subset \tilde{\mathcal{F}}^v \cup \bar{\mathcal{F}}_{s'}^v$, we have that

$$\mu_p(\mathcal{A})\mu_p(\mathcal{B}) \le \mu_p(\tilde{\mathcal{F}}^u \cup \bar{\mathcal{F}}_s^u)\mu_p(\tilde{\mathcal{F}}^v \cup \bar{\mathcal{F}}_{s'}^v) \le f(u, s, p) f(v, s', p) \cdot 1.001^2.$$

Hence, in order to show (3), it suffices to show that

$$g(s, s') := f(u, s, p) f(v, s', p) < 0.99 \left(\mu_p(\mathcal{F}_1^t)\right)^2.$$

Observe that $g(s, s')$ depends on t, p, s and s', but for simplicity we only write the variables s and s'.

Claim 12. *For $s \ge 10$, we have $g(s, s') < 0.99 \left(\mu_p(\mathcal{F}_1^t)\right)^2$.*

Proof. Set $h(\ell, i, p) := p^{-\ell} f(\ell, i, p)$, so that

$$h(\ell, i, p) = \frac{p}{q^{\ell+1}} + \binom{\ell+2i}{i} \frac{\ell+1}{\ell+i+1} (pq)^i (1-\alpha).$$

Since $p^{2t} \le \left(\mu_p(\mathcal{F}_1^t)\right)^2$, it suffices to show that

$$h(u, s, p) h(v, s', p) < 0.99. \tag{6}$$

First, we estimate $h(u, s, p)$. We have that

$$h(u, s, p) \le h(t, s, p) \le h\Big(t, s, \frac{2}{t+3}\Big),$$

where the second inequality holds since p, $1/q$, pq, and $pq(1-\alpha)$ are increasing in p for $0 < p \le 2/(t+3) \le 0.25$. Consequently, since $p/q^{t+1} = \frac{2}{t+3}(\frac{t+3}{t+1})^{t+1} \le 0.073$ for $t \ge 200$, we have

$$\begin{aligned} h\Big(t, s, \frac{2}{t+3}\Big) &\le 0.073 + \binom{t+2s}{s}\Big(\frac{2}{t+3}\Big)^s \le 0.073 + \Big(\frac{e(t+2s)}{s}\frac{2}{t+3}\Big)^s \\ &= 0.073 + \Big(\frac{2e(t+2s)}{s(t+3)}\Big)^s. \end{aligned}$$

Note that $\frac{2e(t+2s)}{s(t+3)} < 1$ if and only if $s > \frac{2et}{t+3-4e}$. Since $s \ge 6 > \frac{2et}{t+3-4e}$ for $t \ge 84$, we have that $\frac{2e(t+2s)}{s(t+3)} < 1$. Also, $\phi(s) := \frac{2e(t+2s)}{s(t+3)}$ is strictly decreasing in s, since $\frac{\phi(s+1)}{\phi(s)} = \frac{(t+2s+2)s}{(t+2s)(s+1)} < 1$. Therefore, for $s \ge 10$,

$$h\Big(t, s, \frac{2}{t+3}\Big) \le 0.073 + \Big(\frac{2e(t+20)}{10(t+3)}\Big)^{10} < 0.08. \tag{7}$$

Next, we estimate $h(v, s', p)$. Similar to the estimation of $h(u, s, p)$, we have that

$$h(v, s', p) \le h(2t, s', p) \le h\Big(2t, s', \frac{2}{t+3}\Big).$$

Consequently, since $p/q^{2t+1} \le 0.53$ for $t \ge 200$, we infer that

$$h\Big(2t, s', \frac{2}{t+3}\Big) \le 0.53 + \binom{2t+2s'}{s'}\Big(\frac{2}{t+3}\Big)^{s'} \le 0.53 + \Big(\frac{4e(t+s')}{s'(t+3)}\Big)^{s'}.$$

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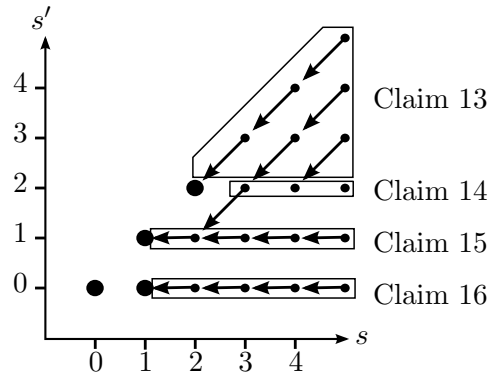

FIGURE 2. $(s_1, s_1') \to (s_2, s_2')$ means $g(s_1, s_1') < g(s_2, s_2')$.

Note that $\frac{4e(t+s')}{s'(t+3)} < 1$ if and only if $s' > \frac{4et}{t+3-4e}$. If $s' \geq 12$, then $s' \geq 12 > \frac{4et}{t+3-4e}$ for $t \geq 84$, and hence, $\frac{4e(t+s')}{s'(t+3)} < 1$. Also, $\psi(s') := \frac{4e(t+s')}{s'(t+3)}$ is strictly decreasing in s', since $\frac{\psi(s'+1)}{\psi(s')} = \frac{(t+s'+1)s'}{(t+s')(s'+1)} < 1$. Hence, for $s' \geq 12$,

$$h\Big(2t, s', \frac{2}{t+3}\Big) \leq 0.53 + \Big(\frac{4e(t+12)}{12(t+3)}\Big)^{12} < 1.2. \tag{8}$$

On the other hand, if $0 \leq s' < 12$, then

$$h\Big(2t, s', \frac{2}{t+3}\Big) \leq 0.53 + \max_{0 \leq s' \leq 11} \left(\binom{2t+2s'}{s'} \Big(\frac{2}{t+3}\Big)^{s'} \right) \leq 11.3. \tag{9}$$

Therefore, we have that

$$h(u,s,p)h(v,s',p) < h\Big(t, s, \frac{2}{t+3}\Big) h\Big(2t, s', \frac{2}{t+3}\Big) \overset{(7),(8),(9)}{<} 0.08 \cdot 11.3 < 0.99,$$

which yields (6). □

4.2. **Intermediate values of s.** To deal with the remaining cases $0 \leq s' \leq s \leq 9$ we show first the monotonicity of $g(s, s')$ on several ranges. See Figure 2 for a schematic of the proof. In Claim 13 we show $g(s, s') < g(s-1, s'-1)$ for values of (s, s') as indicated in the figure. In Claim 14 we show for the indicated cases of (s, s') that $g(s, s') < 0.99 \left(\mu_p(\mathcal{F}_1^t)\right)^2$. In Claims 15 and 16 we show that $g(s, s') < g(s-1, s')$ for the values $s = 0$ and 1 as indicated. The final four values of (s, s'), the larger dots, are dealt with in Section 5.

Claim 13. *For $(s, s') = (3, 2)$ or (s, s') with $3 \leq s' \leq s \leq 9$, we have*

$$g(s, s') < g(s-1, s'-1).$$

Proof. Recall that $g(s, s') = f(u, s, p) f(v, s', p)$ and $g(s-1, s'-1) = f(u, s-1, p) f(v, s'-1, p)$. Hence, it suffices to show that

$$f(u, s, p) < f(u, s-1, p) \tag{10}$$

and

$$f(v, s', p) < f(v, s'-1, p). \tag{11}$$

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First, inequality (10) is equivalent to

$$\binom{u+2s}{s}\frac{pq}{u+s+1} < \binom{u+2s-2}{s-1}\frac{1}{u+s}.$$

We have that

$$\left(\binom{u+2s}{s}\frac{pq}{u+s+1}\right)\Big/\left(\binom{u+2s-2}{s-1}\frac{1}{u+s}\right) \le \frac{(u+2s)(u+2s-1)\cdot 2(t+1)}{s(u+s+1)\cdot(t+3)^2}$$
$$= \frac{2(t+s+s')(t+s+s'-1)(t+1)}{s(t+s'+1)(t+3)^2}.$$

If $(s,s')=(3,2)$, then this is $\frac{2(t+5)(t+4)(t+1)}{3(t+3)^3}$ which is less than one for $t>1$, and if $3\le s'\le s\le 9$, then this is at most $\frac{2(t+18)(t+17)(t+1)}{3(t+4)(t+3)^2}$ which is less than one for $t\ge 54$.

Next, inequality (11) is equivalent to

$$\binom{v+2s'}{s'}\frac{pq}{v+s'+1} < \binom{v+2s'-2}{s'-1}\frac{1}{v+s'}.$$

We have that

$$\left(\binom{v+2s'}{s'}\frac{pq}{v+s'+1}\right)\Big/\left(\binom{v+2s'-2}{s'-1}\frac{1}{v+s'}\right) \le \frac{2(v+2s')(v+2s'-1)(t+1)}{s'(v+s'+1)(t+3)^2}$$
$$= \frac{2(t+s+s')(t+s+s'-1)(t+1)}{s'(t+s+1)(t+3)^2}.$$

If $(s,s')=(3,2)$, then this is $\frac{(t+5)(t+1)}{(t+3)^2}<1$ and if $3\le s'\le s\le 9$, then it is $\frac{2(t+18)(t+17)(t+1)}{3(t+4)(t+3)^2}$ which is less than 1 for $t\ge 54$.

□

Claim 14. *For $3\le s\le 9$ and $s'=2$, we have $g(s,2)<0.99\left(\mu_p(\mathcal{F}_1^t)\right)^2$.*

Proof. Note that $u=t-s+2$ and $v=t+s-2$. Recalling (5), we can write

$$p^s f(u,s,p) = C_1 q^s + C_2 h(s) q^s, \quad \text{and}$$
$$p^{-s} f(v,2,p) = C_3 q^{-s} + C_4 (t+s+2)(t+s-1),$$

where $h(s) := \binom{t+s+2}{s}(t-s+3)p^s$ and

$$C_1=\alpha^{t+3},\quad C_2=\frac{p^{t+2}(1-\alpha)}{t+3},\quad C_3=\alpha^{t-1},\quad \text{and}\quad C_4=\frac{1}{2}p^t q^2(1-\alpha).$$

Note that $C_1,C_2,C_3,C_4>0$ depend only on t and p (and do not depend on s). Multiplying $p^s f(u,s,p)$ and $p^{-s}f(v,1,p)$, we have that

$$\begin{aligned} g(s,2) &= D_1 + D_2 q^s (t+s+2)(t+s-1) + D_3 h(s) + D_4 h(s) q^s (t+s+2)(t+s-1) \\ &\le D_1 + D_2 q^s (t+s+2) 2t + D_3 h(s) + D_4 h(s) q^s (t+s+2)(t+s-1) \qquad (12) \\ &=: g^*(s,2), \end{aligned}$$

where

$$D_1 = C_1C_3 = \alpha^{2t+2}, \qquad D_2 = C_1C_4 = \frac{1}{2}\alpha^{t+3}p^t q^2(1-\alpha),$$
$$D_3 = C_2C_3 = \frac{\alpha^{t-1}p^{t+2}(1-\alpha)}{t+3}, \quad D_4 = C_2C_4 = \frac{(pq)^2 p^{2t}(1-\alpha)^2}{2(t+3)}.$$

Note that $D_1,D_2,D_3,D_4>0$ depend only on t and p (and do not depend on s).

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We claim that $g^*(s,2)$ is strictly decreasing in s for $s \geq 2$. Based on (12), it suffices to show that $a(s) := q^s(t+s+2)$, $h(s)$, and $b(s) := a(s)h(s)(t+s-1)$ are strictly decreasing in s. Indeed, we have that

$$\frac{a(s+1)}{a(s)} = \frac{q(t+s+3)}{t+s+2} \leq \frac{t(t+s+3)}{(t+1)(t+s+2)} < 1$$

and

$$\begin{aligned}\frac{h(s+1)}{h(s)} &= \frac{(t+s+3)(t-s+2)p}{(s+1)(t-s+3)} \leq \frac{2(t+s+3)(t-s+2)}{(s+1)(t-s+3)(t+3)} \\ &< \frac{2(t+s+3)}{(s+1)(t+3)} \leq 1,\end{aligned}$$

where the first inequality follows from $p \leq 2/(t+3)$ and the last inequality holds for $s \geq 2$. Finally,

$$\frac{b(s+1)}{b(s)} < \frac{2(t+s+3)}{(s+1)(t+3)} \cdot \frac{t+s}{t+s-1} < \frac{2(t+12)}{3(t+3)} \cdot \frac{t+1}{t} < 1,$$

where the second inequality holds because $2 \leq s \leq 9$ and the last inequality holds for $t \geq 19$.

Therefore, it suffices to show that $g^*(3,2) < 0.99\left(\mu_p(\mathcal{F}_1^t)\right)^2$. We have that

$$\frac{g^*(3,2)}{\left(\mu_p(\mathcal{F}_1^t)\right)^2} < \frac{g^*(3,2)}{(t+2)^2 p^{2t+2} q^2}.$$

Since $1/e^2 \leq q^t \leq 1/2$, we infer that

$$\begin{aligned}\frac{g^*(3,2)}{\left(\mu_p(\mathcal{F}_1^t)\right)^2} <& \frac{e^4}{(t+2)^2 q^4} + \frac{e^2 p(1-\alpha)(t+5)t}{(t+2)^2} + \frac{e^2 p^2 (1-\alpha)(t+5)(t+4)t}{6(t+2)^2 q} \\ &+ \frac{p^3 q^3 (1-\alpha)^2 (t+5)^2 (t+4)t}{12(t+2)}. \end{aligned} \tag{13}$$

Since $1/q, p(1-\alpha), p^2(1-\alpha), p^3q^3(1-\alpha)^2$ are increasing in p for $p < 0.2$, we have that (13) is maximized when $p = 2/(t+3)$. One can check that (13) with $p = 2/(t+3)$ is at most 0.99 for $t \geq 48$. □

Claim 15. *For $s \geq 2$ and $s' = 1$, we have $g(s,1) < g(1,1)$.*

Proof. As in Claim 14, we can write,

$$\begin{aligned} p^s f(u,s,p) &= C_1 q^s + C_2 h(s) q^s, \quad \text{and} \\ p^{-s} f(v,1,p) &= C_3 q^{-s} + C_4 (t+s), \end{aligned}$$

where $h(s) := \binom{t+s+1}{s}(t-s+2)p^s$. As now, $u = t-1+s$ and $v = t = s-1$, the values $C_1, C_2, C_3, C_4 > 0$ are slightly different from in Claim 14, but still they depend only on t and p (and not on s). Multiplying $p^s f(u,s,p)$ and $p^{-s} f(v,1,p)$, we have that

$$g(s,1) = D_1 + D_2 q^s (t+s) + D_3 h(s) + D_4 h(s) q^s (t+s), \tag{14}$$

where $D_1, D_2, D_3, D_4 > 0$ depend only on t and p.

We claim that $g(s,1)$ is strictly decreasing in s for $s \geq 1$. By (14), it suffices to show that $q^s(t+s)$ and $h(s)$ are strictly decreasing in s. First, $q^s(t+s)$ is strictly decreasing in s since

$$\frac{q^{s+1}(t+s+1)}{q^s(t+s)} = \frac{q(t+s+1)}{(t+s)} \leq \frac{t(t+s+1)}{(t+1)(t+s)} < 1,$$

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where the first inequality follows from $q \le t/(t+1)$ and the last inequality holds for $s \ge 1$. Next, $h(s)$ is strictly decreasing in s since

$$\frac{h(s+1)}{h(s)} \le \frac{2(t+s+2)(t-s+1)}{(s+1)(t-s+2)(t+3)} < \frac{2(t+s+2)}{(s+1)(t+3)} \le 1,$$

where the first inequality follows from $p \le 2/(t+3)$ and the last inequality follows from $s \ge 1$. □

Claim 16. *For $s \ge 2$ and $s' = 0$, we have $g(s,0) < g(1,0)$.*

Proof. Again, noting this time that $u = t - s$ and $v = t + s$, we write

$$p^s f(u,s,p) = C_1 q^s + C_2 h(s) q^s, \quad \text{and}$$
$$p^{-s} f(v,0,p) = C_3 q^{-s} + C_4,$$

where $h(s) := \binom{t+s}{s}(t-s+1)p^s$, and $C_1, C_2, C_3, C_4 > 0$ (different from above) depend only on t and p. Multiplying $p^s f(u,s,p)$ and $p^{-s} f(v,0,p)$, we have that

$$g(s,0) = D_1 + D_2 q^s + D_3 h(s) + D_4 h(s) q^s, \tag{15}$$

where $D_1, D_2, D_3, D_4 > 0$ depend only on t and p.

We claim that $g(s,0)$ is strictly decreasing in s for $s \ge 1$. By (15), it suffices to show that $h(s)$ is strictly decreasing in s. Indeed,

$$\frac{h(s+1)}{h(s)} \le \frac{2(t+s+1)(t-s)}{(s+1)(t-s+1)(t+3)} < \frac{2(t+s+1)}{(s+1)(t+3)} \le 1,$$

where the first inequality follows from $p \le 2/(t+3)$ and the last inequality holds for $s \ge 1$. □

Referring to our outline of the proof preceding Claim 13, (and Figure 2), Claims 12–16 imply the following corollary.

Corollary 17. *If $\mu_p(\mathcal{A})\mu_p(\mathcal{B}) \le \left(\mu_p(\mathcal{F}_1^t)\right)^2$ holds for*

$$(s,s') = (0,0), (1,0), (1,1), (2,2),$$

then, for all (s,s') other than $(0,0), (1,0), (1,1), (2,2)$,

$$\mu_p(\mathcal{A})\mu_p(\mathcal{B}) < \left(\mu_p(\mathcal{F}_1^t)\right)^2.$$

5. Remaining cases

5.1. **Definitions.** We introduce several definitions and notation. For $A \subset [n]$, let $(A)_i$ be the i-th smallest element of A. For $A, B \subset [n]$, we say that A *shifts to* B, denoted by

$$A \to B,$$

if $|A| \le |B|$ and $(A)_i \ge (B)_i$ for each $i \le |A|$. In other words, as walks on a two-dimensional grid, each edge of the walk B is not contained in the area to the right of the walk A. For example, $\{2,4,6\} \to \{1,4,5,7\}$.

Fact 18 ([7], Fact 2.8)**.** *Let $\mathcal{F}$ be a shifted, inclusion maximal family in $2^{[n]}$. If $F \in \mathcal{F}$ and $F \to F'$, then $F' \in \mathcal{F}$.*

This immediately implies the follows.

Fact 19. *Let $\mathcal{F}$ be a shifted, inclusion maximal family in $2^{[n]}$. If $F' \notin \mathcal{F}$, then every $F \in \mathcal{F}$ satisfies $F \not\to F'$.*

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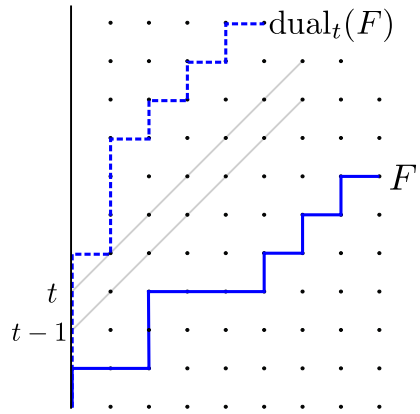

FIGURE 3. The walk dual$_t(F)$ of F with respect to t

For $t \in [n]$ and $F \subset [n]$, the *dual* of F with respect to t is defined by

$$\mathrm{dual}_t(F) := [(F)_t - 2] \cup ([n] \setminus F).$$

Viewed as walks on a two-dimensional grid, the walk dual$_t(F)$ is obtained by reflecting F across the line $y = x + (t-1)$ and ignoring the part $x < 0$. (See Figure 3.)

Fact 20 ([7], Fact 2.9)**.** *Let $\mathcal{A}$ and $\mathcal{B}$ be cross t-intersecting families. If $A \in \mathcal{A}$, then* dual$_t(A) \notin \mathcal{B}$.

5.2. **The case** $(s, s') = (1, 0)$**.**

Lemma 21. *For $(s, s') = (1, 0)$, we have $\mu_p(\mathcal{A})\mu_p(\mathcal{B}) < \big(\mu_p(\mathcal{F}_1^t)\big)^2$.*

In our proof of Lemma 21, we avoid several tedious calculations by appealing to estimations done in the proof of Lemma 3.5 in [7]. Many of these estimations contain a value $\epsilon > 0$ that can be taken arbitrarily small by assuming that n is sufficiently large, (where how large 'sufficiently large' is usually depends on t). We will frequently do this; recall from our setup that it is permitted.

Proof. Note that $u = t - 1$ and $v = t + 1$. For integers $\ell, i \geq 1$ and $s \geq 0$, let

$$\begin{aligned} D_s^\ell(i) :=& [\ell - 1] \cup \{\ell + s, \ell + s + 1, \dots, \ell + 2s\} \\ & \cup \{\ell + 2s + i + 2k \in [n] : k = 1, 2, \dots\} \in \dot{\mathcal{F}}^\ell \cap \mathcal{F}_s^\ell. \end{aligned} \tag{16}$$

Note that $D_s^\ell(i) = D_s^\ell(n - \ell - 2s - 1)$ for $i \geq n - \ell - 2s - 1$, and hence, we assume that

$$1 \leq i \leq n - \ell - 2s - 1. \tag{17}$$

In particular, consider the following walks. For $1 \leq i \leq n - t - 2$, let

$$D_1^{t-1}(i) = [t-2] \cup \{t, t+1\} \cup \{t + 1 + i + 2k \in [n] : k = 1, 2, \dots\} \in \dot{\mathcal{F}}^{t-1} \cap \mathcal{F}_1^{t-1}.$$

For $1 \leq j \leq n - t - 2$, let

$$D_0^{t+1}(j) = [t+1] \cup \{t + 1 + j + 2k \in [n] : k = 1, 2, \dots\} \in \dot{\mathcal{F}}^{t+1} \cap \mathcal{F}_0^{t+1}.$$

The walks $D_1^{t-1}(i)$ and $D_0^{t+1}(j)$ are denoted by $D_i^{\mathcal{A}}$ and $D_j^{\mathcal{B}}$ in [7]. (They are depicted in Figure 1 of [7].)

One can easily see that any walk in $\dot{\mathcal{B}}$ shifts to $D_0^{t+1}(1)$, and as $s = 1$ any walk in $\dot{\mathcal{A}}$ shifts to $D_1^{t-1}(1)$. Thus by Fact 18 and the fact that $\dot{\mathcal{A}} \neq \emptyset$ and $\dot{\mathcal{B}} \neq \emptyset$, we have that $D_1^{t-1}(1) \in \mathcal{A}$ and

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$D_0^{t+1}(1) \in \mathcal{B}$. So $\{i : D_1^{t-1}(i) \in \mathcal{A}\}$ and $\{j : D_0^{t+1}(j) \in \mathcal{B}\}$ are not empty, and the following values are well defined:

$$I := \max\{i : D_1^{t-1}(i) \in \mathcal{A}\} \quad \text{and} \quad J := \max\{j : D_0^{t+1}(j) \in \mathcal{B}\}.$$

We consider the following three cases separately.

• **Case 1:** *Suppose that $I \geq 2$ and $J \geq 2$.* Inequalities (11) and (12) in [7] give that

$$\begin{aligned}\mu_p(\mathcal{A}) = \mu_p(\tilde{\mathcal{A}}) + \mu_p(\mathcal{A}_1) &\leq \alpha^{t+J-1} + p^t + tp^t q(1+\epsilon) \\ &\leq \alpha^{t+1} + p^t + tp^t q(1+\epsilon) \leq 2\alpha^{t+1} + p^t + tp^t q,\end{aligned}$$

where by taking n sufficiently large we may assume that $\epsilon > 0$ is such that $tp^t q\epsilon \leq \alpha^{t+1}$. Inequalities (14) and (15) in [7] imply that

$$\begin{aligned}\mu_p(\mathcal{B}) = \mu_p(\tilde{\mathcal{B}}) + \mu_p(\mathcal{B}_0) &\leq \alpha^{t+1+I} + p^{t+1}(1+\epsilon) \\ &\leq \alpha^{t+3} + p^{t+1}(1+\epsilon) \leq 2\alpha^{t+3} + p^{t+1},\end{aligned}$$

where by taking n sufficiently large we may assume that $\epsilon > 0$ is such that $p^{t+1}\epsilon \leq \alpha^{t+3}$.

Thus

$$\frac{\mu_p(\mathcal{A})\mu_p(\mathcal{B})}{(\mu(\mathcal{F}_1^t))^2} \leq \frac{\left(\frac{2e^2p}{q} + 1 + tq\right)\left(\frac{2e^2p^2}{q^3} + 1\right)}{pq^2(t+2)^2} \leq \frac{\left(\frac{4e^2}{t+1} + 1 + \frac{t^2}{t+1}\right)\left(\frac{8e^2(t+3)}{(t+1)^3} + 1\right)}{(t+2)^2t^2/(t+1)^3} < 1,$$

where the first inequality follows from (4) and the second inequality holds since $p, 1/q$, and pq^2 are increasing in p for $p \leq 0.2$, and the last inequality holds for $t \geq 100$. Therefore, we have

$$\mu_p(\mathcal{A})\mu_p(\mathcal{B}) < \left(\mu_p(\mathcal{F}_1^t)\right)^2.$$

• **Case 2:** *Suppose that $I = 1$.* As $\mathcal{B} \subset \mathcal{F}^{t+1}$, we have by Lemma 2(i) that $\mu_p(\mathcal{B}) \leq \alpha^{t+1}$, while the inequalities on lines 7 and 16 of page 227 of [7] yield that

$$\mu_p(\mathcal{A}) \leq \alpha^t + p^t\Big((1-\alpha)\left(tq - (t-1)q^3\right) + tq\epsilon\Big).$$

By taking n sufficiently large we can assume that ϵ satisfies $\epsilon < \frac{2(t-2)}{t(t+1)(t+3)}$ and $tq\epsilon < p^2(t-1)$. So,

$$\begin{aligned}(1-\alpha)(tq - (t-1)q^3) + tq\epsilon &< \left(t(1-p) - (t-1)(1-p)^3\right) + tq\epsilon \\ &< \left(1 + p(t-2) - p^2(t-1)\right) + tq\epsilon \\ &< 1 + p(t-2).\end{aligned}$$

Consequently, $\mu(\mathcal{A}) < p^t\left(\frac{1}{q^t} + 1 + p(t-2)\right) < p^t\left(e^2 + 1 + p(t-2)\right)$, where the second inequality follows from $q^t > 1/e^2$.

Together this gives

$$\frac{\mu_p(\mathcal{A})\mu_p(\mathcal{B})}{(\mu_p(\mathcal{F}_1^t))^2} < \frac{p^t\left(e^2 + 1 + p(t-2)\right)\frac{p^{t+1}}{q^{t+1}}}{(t+2)^2p^{2t+2}q^2} < \frac{e^2\left(e^2 + 1 + p(t-2)\right)}{(t+2)^2pq^3}.$$

Since p and pq^3 are increasing in p for $p \leq 0.2$, we infer that

$$\frac{\mu_p(\mathcal{A})\mu_p(\mathcal{B})}{(\mu_p(\mathcal{F}_1^t))^2} < \frac{e^2\left(e^2 + 1 + \frac{2(t-2)}{t+3}\right)}{(t+2)^2\frac{t^3}{(t+1)^4}} < 0.99,$$

where the last inequality holds for $t \geq 92$.

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• **Case 3:** *Suppose that* $J = 1$. Then $\mu_p(\mathcal{A}) \le \alpha^{t-1}$, and the inequalities in lines 12 and 17 in page 228 of [7] imply that

$$\mu_p(\mathcal{B}) \le \alpha^{t+2} + p^{t+1}\left(1 - \alpha + \epsilon - q^2(1-\alpha)\right).$$

Taking sufficiently large n we can ensure that $\epsilon \le (\frac{1}{t+1})^2 \le p^2$. So

$$\begin{aligned} 1 - \alpha + \epsilon - q^2(1-\alpha) &= 1 - (1-p)(1-2p) - \frac{p}{1-p} + \epsilon \\ &= 1 + 3p - 2p^2 - \frac{1}{q} + \epsilon < 1 + 3p - \frac{1}{q}. \end{aligned}$$

Consequently, $\mu(\mathcal{B}) < p^{t+1}\left(\frac{p}{q^{t+2}} + 1 + 3p - \frac{1}{q}\right) < p^{t+1}\left(\frac{e^2 p}{q^2} + 1 + 3p - \frac{1}{q}\right)$, where the second inequality follows from $q^t > 1/e^2$.

We have that

$$\begin{aligned} \frac{\mu_p(\mathcal{A})\mu_p(\mathcal{B})}{\left(\mu_p(\mathcal{F}_1^t)\right)^2} &< \frac{\frac{p^{t-1}}{q^{t-1}} p^{t+1}\left(\frac{e^2 p}{q^2} + 1 + 3p - \frac{1}{q}\right)}{(t+2)^2 p^{2t+2} q^2} < \frac{e^2\left(\frac{e^2 p}{q^2} + 1 + 3p - \frac{1}{q}\right)}{(t+2)^2 p^2 q} \\ &\le \frac{e^2\left(\frac{2e^2(t+3)}{(t+1)^2} + 1 + \frac{3}{t+1} - \frac{t+1}{t}\right)}{(t+2)^2 t/(t+1)^3} < 0.99, \end{aligned}$$

where the last inequality holds for $t \ge 125$. □

5.3. **Extremal cases.** Finally, we consider the cases $(s, s') = (0,0)$, $(1,1)$, $(2,2)$.

Lemma 22. *For* $(s,s) = (0,0), (1,1), (2,2)$, *we have*

$$\mu_p(\mathcal{A})\mu_p(\mathcal{B}) \le \left(\mu_p(\mathcal{F}_1^t)\right)^2. \tag{18}$$

Moreover, equality holds if and only if one of the following holds:

(i) $\mathcal{A} = \mathcal{B} = \mathcal{F}_0^t$ *and* $p = \dfrac{1}{t+1}$,

(ii) $\mathcal{A} = \mathcal{B} = \mathcal{F}_1^t$ *and* $\dfrac{1}{t+1} \le p \le \dfrac{2}{t+3}$,

(iii) $\mathcal{A} = \mathcal{B} = \mathcal{F}_2^t$ *and* $p = \dfrac{2}{t+3}$.

Proof. In these cases, we have $u = v = t$. Recalling (16) and (17), we let

$$\begin{aligned} D_s^t(i) :=& [t-1] \cup \{t+s, t+s+1, \dots, t+2s\} \\ &\cup \{t+2s+i+2k \in [n] : k = 1, 2, \dots\} \in \dot{\mathcal{F}}^t \cap \mathcal{F}_s^t \end{aligned}$$

for $1 \le i \le n - t - 2s - 1 =: i_{\max}$.

In order to define

$$I = \max\{i \,:\, D_s^t(i) \in \mathcal{A}\} \quad \text{and} \quad J = \max\{i \,:\, D_s^t(i) \in \mathcal{B}\},$$

the sets $\{i : D_s^t(i) \in \mathcal{A}\}$ and $\{i : D_s^t(i) \in \mathcal{B}\}$ should not be empty. Hence, we consider the following two cases separately:

- **Case I:** $D_s^t(1) \in \mathcal{A}$ and $D_s^t(1) \in \mathcal{B}$.
- **Case II:** $D_s^t(1) \notin \mathcal{A}$ or $D_s^t(1) \notin \mathcal{B}$

As we observed in the proof of Lemma 21, Case I holds if $s = 0$ or 1.

• **Case I:** *Suppose that* $D_s^t(1) \in \mathcal{A}$ *and* $D_s^t(1) \in \mathcal{B}$.

First, we suppose that $I = J = i_{\max}$. Since $D_s^t(i_{\max}) \in \mathcal{A}$, Fact 20 gives that

$$\mathrm{dual}_t\left(D_s^t(i_{\max})\right) = [n] \setminus \{t+s, t+s+1, \dots, t+2s\}$$

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is not contained in $\mathcal{B}$. Consequently, Fact 19 gives that each walk $B \in \mathcal{B}$ satisfies $B \not\to \mathrm{dual}_t\left(D_s^t(i_{\max})\right)$. Hence, $\mathcal{B} \subset \mathcal{F}_s^t$ holds. Similarly, $J = i_{\max}$ implies $\mathcal{A} \subset \mathcal{F}_s^t$. Therefore, we have

$$\mu_p(\mathcal{A})\mu_p(\mathcal{B}) \le \left(\mu_p(\mathcal{F}_s^t)\right)^2$$

with equality holding iff $\mathcal{A} = \mathcal{B} = \mathcal{F}_s^t$. This together with $1/(t+1) \le p \le 2/(t+3)$ implies (18) and (22).

Therefore, we can assume that $I \ne i_{\max}$ or $J \ne i_{\max}$. Without loss of generality, let $I \ne i_{\max}$. The following holds for every $s \ge 0$ (not only for $0 \le s \le 2$).

Claim 23. *If $I \ne i_{\max}$ and $0 \le s < \infty$, then*

$$\mu_p(\mathcal{F}_s^t \setminus \mathcal{A}) \ge \binom{t+s-1}{s} p^{t+s} q^{s+I+1}(1-\alpha) \tag{19}$$

and

$$\mu_p(\mathcal{B} \setminus \mathcal{F}_s^t) \le \alpha^{t+I}. \tag{20}$$

Proof. First, we show (19). Consider a walk W that hits $(s, t+s)$ and satisfies $W \to D_s^t(I+1)$. Since $D_s^t(I+1) \notin \mathcal{A}$, Fact 19 gives $W \in \mathcal{F}_s^t \setminus \mathcal{A}$. Also, W must hit $Q_1 = (s, t-1)$ and $Q_2 = (s+I+1, t+s)$. The number of walks from $(0,0)$ to Q_1 is $\binom{t+s-1}{s}$, then there is the unique walk from Q_1 to Q_2 which hits $(s, t+s)$. So the weight of the family of all such walks is $\binom{t+s-1}{s} p^{t+s} q^{s+I+1}$. To satisfy $W \to D_{I+1}$, the walk W must not hit the line $y = x + (t-I)$. By Lemma 2(i), this happens with probability at least $1-\alpha$, which yields (19).

Next, we show (20). Since $D_s^t(I) \in \mathcal{A}$, we have that $\mathrm{dual}_t\left(D_s^t(I)\right) \notin \mathcal{B}$, and hence, each walk $B \in \mathcal{B}$ must hit $(0, t+s), (1, t+s), \ldots, (s, t+s)$, or $y = x + (t+I)$. Note that each walk hitting $(0, t+s), (1, t+s), \ldots,$ or $(s, t+s)$ is contained in $\mathcal{F}_s^t$. Thus, each walk $B \in \mathcal{B} \setminus \mathcal{F}_s^t$ hits $y = x + (t+I)$. Lemma 2(i) gives (20). □

Claim 23 together with $0 \le s \le 2$ implies the following.

Corollary 24. *If $I \ne i_{\max}$ and $0 \le s \le 2$, then $\mu_p(\mathcal{B} \setminus \mathcal{F}_s^t) < 0.99\mu_p(\mathcal{F}_s^t \setminus \mathcal{A})$.*

Proof. Inequalities (19) and (20) give that

$$\begin{aligned}\frac{\mu_p(\mathcal{F}_s^t \setminus \mathcal{A})}{\mu_p(\mathcal{B} \setminus \mathcal{F}_s^t)} &\ge \frac{\binom{t+s-1}{s} p^{t+s} q^{s+I+1}(1-\alpha)}{\alpha^{t+I}} = \binom{t+s-1}{s} p^s q^{t+s+1} \left(\frac{q^2}{p}\right)^I (1-\alpha) \\ &\ge \binom{t+s-1}{s} p^s q^{t+s} \left(\frac{q^2}{p}\right)(q-p) = \binom{t+s-1}{s} p^{s-1} q^{t+s+2}(q-p),\end{aligned}$$

where the second inequality holds since $q^2/p > 1$ for $p < 0.38$. Since $1/(t+1) \le p \le 2/(t+3)$, one can easily check that

$$\binom{t+s-1}{s} p^{s-1} q^{t+s+2}(q-p) > 1.02$$

if $s = 0$ and $t \ge 17$, or if $s = 1$ and $t \ge 12$, or if $s = 2$ and $t \ge 22$. □

On the other hand, we also claim that

$$\mu_p(\mathcal{A} \setminus \mathcal{F}_s^t) < 0.99\mu_p(\mathcal{F}_s^t \setminus \mathcal{B}). \tag{21}$$

Indeed, if $J \ne i_{\max}$, then Corollary 24 gives (21). Otherwise, $J = i_{\max}$, and hence, $\mathcal{A} \subset \mathcal{F}_s^t$. Thus, (21) trivially holds.

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We infer that

$$\begin{aligned}\mu_p(\mathcal{A})+\mu_p(\mathcal{B}) &= \big(\mu_p(\mathcal{A}\cap\mathcal{F}_s^t)+\mu_p(\mathcal{A}\setminus\mathcal{F}_s^t)\big)+\big(\mu_p(\mathcal{B}\cap\mathcal{F}_s^t)+\mu_p(\mathcal{B}\setminus\mathcal{F}_s^t)\big)\\ &< \big(\mu_p(\mathcal{A}\cap\mathcal{F}_s^t)+0.99\mu_p(\mathcal{F}_s^t\setminus\mathcal{A})\big)+\big(\mu_p(\mathcal{B}\cap\mathcal{F}_s^t)+0.99\mu_p(\mathcal{F}_s^t\setminus\mathcal{B})\big)\\ &< 2\mu_p(\mathcal{F}_s^t),\end{aligned}$$

where the inequality follows from Corollary 24 and (21). Therefore,

$$\sqrt{\mu_p(\mathcal{A})\mu_p(\mathcal{B})}\le\frac{\mu_p(\mathcal{A})+\mu_p(\mathcal{B})}{2}<\mu_p(\mathcal{F}_s^t),$$

which gives (18) without equality.

• **Case II:** *Suppose that $D_s^t(1)\notin\mathcal{A}$ or $D_s^t(1)\notin\mathcal{B}$.*

As Case I holds if $s=0,1$, we may assume that $s=2$. Also, without loss of generality, we let $D_s^t(1)\notin\mathcal{A}$, so every $A\in\mathcal{A}$ satisfies $A\not\to D_s^t(1)$.

For $1\le i\le n-t-5$, let

$$E(i):=[t-1]\cup\{t+1,t+3,t+4\}\cup\{t+4+i+2j\in[n]:j=1,2,\ldots\}.$$

For any $A\in\dot{\mathcal{A}}\neq\emptyset$, we have $A\to E(1)$, and hence, Fact 18 gives that $E(1)\in\mathcal{A}$. Since $\{i:E(i)\in\mathcal{A}\}\neq\emptyset$, the number

$$K:=\max\{i:E(i)\in\mathcal{A}\}$$

is well-defined.

Let $A\in\dot{\mathcal{A}}$. The walk A must hit $(2,t+2)$ without hitting $(0,t)$ or $(1,t+1)$. Also, since $D_s^t(1)\notin\mathcal{A}$, the walk A must hit $(1,t)$. The weight of the family of all such walks is $tp^{t+2}q^2$. From $(2,t+2)$, the walk A moves to the right and hits $(3,t+2)$. Then it must not hit the line $y=x+t$. Lemma 3 implies that this happens with probability less than $q(1-\alpha+\epsilon)$ where $\epsilon\to0$ as n tends to ∞. Let n be sufficiently large that $\epsilon\le\alpha$. Then, we have that

$$\mu_p(\dot{\mathcal{A}})<tp^{t+2}q^3(1-\alpha+\epsilon)\le tp^{t+2}q^3.$$

For $\ddot{\mathcal{A}}$ and $\tilde{\mathcal{A}}$ we use the trivial bounds $\mu_p(\ddot{\mathcal{A}})\le\alpha^{t+1}$ and $\mu_p(\tilde{\mathcal{A}})\le\alpha^{t+1}$ from Lemma 2. Consequently we have

$$\mu_p(\mathcal{A})=\mu_p(\dot{\mathcal{A}})+\mu_p(\ddot{\mathcal{A}})+\mu_p(\tilde{\mathcal{A}})<tp^{t+2}q^3+2\alpha^{t+1}. \tag{22}$$

On the other hand,

$$\begin{aligned}\mathrm{dual}_t(E(K)) &= [t]\cup\{t+2\}\cup\{t+5,\ldots,t+5+K\}\\ &\quad\cup\{t+5+2j\in[n]:j=1,2,\ldots\},\end{aligned}$$

Since $\mathrm{dual}_t(E(K))\notin\mathcal{B}$, every walk $B\in\mathcal{B}$ must hit one of $(0,t+1)$, $(1,t+2)$, $(2,t+2)$, or the line $y=x+t+K$. Thus we have

$$\begin{aligned}\mu_p(\mathcal{B}) &\le p^{t+1}+(t+1)p^{t+2}q+\left((t+1)+\binom{t+2}{2}\right)p^{t+2}q^2+\alpha^{t+K}\\ &\le\left(1+(t+1)pq+\frac{(t+1)(t+4)}{2}pq^2\right)p^{t+1}+\alpha^{t+1}.\end{aligned} \tag{23}$$

Inequalities (22) and (23) give that

$$\begin{aligned}\frac{\mu_p(\mathcal{A})\mu_p(\mathcal{B})}{(\mu_p(\mathcal{F}_1^t))^2} &\le\frac{\left(tp^{t+2}q^3+2\alpha^{t+1}\right)\left(1+(t+1)pq+\frac{(t+1)(t+4)}{2}pq^2+1/q^{t+1}\right)p^{t+1}}{((t+2)p^{t+1}q)^2}\\ &\le\frac{\left(tpq^3+\frac{2e^2}{q}\right)\left(1+(t+1)pq+\frac{(t+1)(t+4)}{2}pq^2+\frac{e^2}{q}\right)}{(t+2)^2q^2},\end{aligned} \tag{24}$$

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where the second inequality follows from (4). Since pq, pq^2, pq^3 and $1/q$ are increasing in p for $p \leq 0.2$, expression (24) is maximized when $p = 2/(t+3)$. One can check that (24) with $p = 2/(t+3)$ is at most 0.99 for $t \geq 180$. Therefore, for $t \geq 180$,

$$\mu_p(\mathcal{A})\mu_p(\mathcal{B}) < 0.99 \left(\mu_p(\mathcal{F}_1^t)\right)^2, \tag{25}$$

which completes our proof of Lemma 22. □

We have proved the inequality (3) under Assumption 7. The uniqueness of the optimal families in Theorem 1 now follows from Lemma 6. This completes the proof of Theorem 1.

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